Proposal for the provision of EPC and financing services of virtual net metering PV system for Four Seasons Astir Palace Hotel Athens

ZEB AEEY

Energy Services Company SA

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Zero Energy Building

1. COMPANY PROFILE – OPERATIONS

ZEB SA is one of the first Greek companies providing energy services (ESCOs).

Since 2020 Alpha Bank participates in the ownership structure of ZEB, through its VC. Specifically, in the field of photovoltaic investments ZEB provides various services for the bank as its formal advisor. Furthermore, ZEB can provide project financing though its cooperation with a green fund.

- Pioneers in designing, developing and implementing ENERGY PERFORMANCE CONTRACTS (EPC) in the domestic market
- Among leaders in Engineering-procurement-contracting of photovoltaic installations in Greece
- Specialized group of energy engineers, auditors and analysts in energy design and energy management of buildings
- Certified staff with international experience in managing ESCO projects
- Experience in the Application of International Standards and Practice of Energy Management (IPMVP, Ashrae, EN, ISO)

Steadily orientated in creating long lasting principals and sustainable development and through our experience in dealing with complex building issues, in ZEB we desire to support the national effort for achieving energy and environmental targets through projects we undertake, operating at the same time as consultants in organizing and applying national programs.

In ZEB SA we design services which aim at the best possible energy performance private and public organizations through energy conservation, use of alternative power sources and optimum use of human and financial resources.

We define, design and implement energy saving interventions in building and industrial installations (photovoltaics, building shell, E/M installations & equipment, energy management etc) assuring concurrently project implementation with third-party financing. We guarantee specific energy savings, from which each project is repaid individually. Therefore, proposed energy savings projects are feasible solutions for the final client.

Each project is paid according to specific energy savings target which are guaranteed. In this way, the proposed energy saving projects are solutions feasible for the final consumer.

2015 ENVIRONMENTAL AWARDS

ΠΙΣΤΟΠΟΙΗΤΙΚΟ

Η ZERO ENERGY BUILDING ZEB AEEY

στα πλαίσια των βραβείων ***Environmental Awards 2015***, τα οποία διοργάνωσε η εταιρεία ***Boussias Communications***, διακρίθηκε με **BRONZE** βραβείο στην κατηγορία " **Energy Services Companies & Consultants**" για το **έργο**
"ZEB AEEY ως εταιρεία ενεργειακών υπηρεσιών ESCO " .

Μιχάλης Κ. Μπούσιας

Εκδότης

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2016 ENVIRONMENTAL AWARDS

ΠΙΣΤΟΠΟΙΗΤΙΚΟ

Η ZERO ENERGY BUILDING ZEB AEEY

στα πλαίσια των βραβείων ***Environmental Awards 2016***, τα οποία διοργάνωσε η εταιρεία ***Boussias Communications***,
διακρίθηκε με **GOLD** βραβείο στην κατηγορία " **Πιστοποίηση ενεργειακής απόδοσης**"
για το **έργο**
ΕΦΑΡΜΟΓΗ ΣΥΣΤΗΜΑΤΟΣ ΜΕΤΡΗΣΗΣ & ΕΠΑΛΗΘΕΥΣΗΣ ΕΞΟΙΚΟΝΟΜΗΣΗΣ IPVMP.

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Zero Energy Building

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Redex

LABOLAGET 360° FACILITY MANAGEMENT

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air bp

ΜΟΡΝΟΣ

SILGAN METAL PACKAGING

Boehringer Ingelheim

NOVAL PROPERTY

vitex

whitestone engineers & contractors

ΕΡΤ

ΟΜΙΛΟΣ ΝΗΡΕΥΣ

ΕΛΛΗΝΙΚΗ ΕΤΑΙΡΕΙΑ ΣΥΜΜΕΤΟΧΩΝ ΚΑΙ ΠΕΡΙΟΥΣΙΑΣ

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ΣΤΑΣΥ ΣΤΑΘΕΡΕΣ ΣΥΓΚΟΙΝΩΝΙΕΣ Α.Ε.

ελιν

ΕΒΓΑ

ΔΗΜΟΣ ΒΟΛΟΥ

ΕΛΛΗΝΙΚΗ ΑΕΡΟΠΟΡΙΚΗ ΒΙΟΜΗΧΑΝΙΑ Α.Ε.
EN 9100:2016 / AS 9100D
ISO 9001:2015

Ελληνική Δημοκρατία ΠΕΡΙΦΕΡΕΙΑ ΘΕΣΣΑΛΙΑΣ

ΔΗΜΟΣ ΠΕΙΡΑΙΑ

2. INTRODUCTION

In the new business environment requiring a lower **environmental footprint of companies, smart technologies and the proper management of energy resources**, investors, employees, consumers and the pubic require from companies to take these factors into account.

The Environmental, Social and Corporate Governance (ESG) criteria are an increasingly popular company evaluation tool used by all stakeholders to select partnerships, investments and products that match their values. Environmental criteria may include a company's energy use, waste, pollution, conservation of natural resources and animal treatment. The criteria can also be used to assess any environmental risks that a company may face and how the company manages these risks. In addition, investors use the criteria in order to avoid financial risks due to environmental or other corporate practices.

Today, Four Seasons Astir Palace Hotel Athens is able to lower its environmental footprint by **reducing its carbon footprint while achieving large energy cost reductions**. This can be applied though the practice of net-metering (compensating energy consumption with photovoltaic production on site). In the case of Four Seasons Astir Palace Hotel Athens, due to a number of reasons (lack of space and aesthetic) net-metering cannot be applied on site. Therefore, the solution of **virtual net-metering**, can be promoted where **the energy consumed can compensated by the energy produced in a remote site**. The vehicle through which this compensation can be applied is an **Energy Community**. The Energy community is a non-profit organization of five members (VAT registries) with Four Seasons Astir Palace Hotel Athens as a member.

In the next chapters, the reduction of energy cost and relevant carbon footprint reduction through virtual energy net metering by establishing an Energy community is presented for three financing options.

3. EXECUTIVE SUMMARY

This proposal concerns the provision of turnkey construction services, commission, maintenance and monitoring of operation for a photovoltaic park. The photovoltaic system will be connected to the network under the regime of virtual net metering energy compensation as defined by the latest provisions of the Greek regulatory environment.

The park will provide the following benefits for the company:

- Electricity consumption **carbon footprint reduction (76%)**
- Electricity **cost reduction (51%)**

For the compensation of the consumed electricity with the generation of the park an energy community needs to be established as a vehicle of performing the virtual net-metering of the produced with the consumed energy.

Our proposal involves three options of financing with different terms. In specific, the third option presents an eight-year energy performance contract with low capex participation from the client.

The results for the client are presented in the following table:

Summary results	Option 1 - Self financing	Option 2 – 70% Debt	Option 3 – Energy Performance Contract
CAPEX participation	2.135.000 €	640.000 €	170.000 €
Payback	8 years	5 years	10 years
Payout price (year 8)	0 €	0 €	170.000 €
NPV year 8	-314.451 €	219.473 €	-74.073 €
IRR (25-year)	12,2%	21,7%	21,9%
NPV (5%, 25-year)	4.548.925 €	1.712.977 €	1.284.094 €

This proposal is an **initial estimation referring to budgeting assumptions**. The actual figures will be renewed after the detailed analysis that will follow after acceptance.

4. VIRTUAL NET-METERING DESCRIPTION

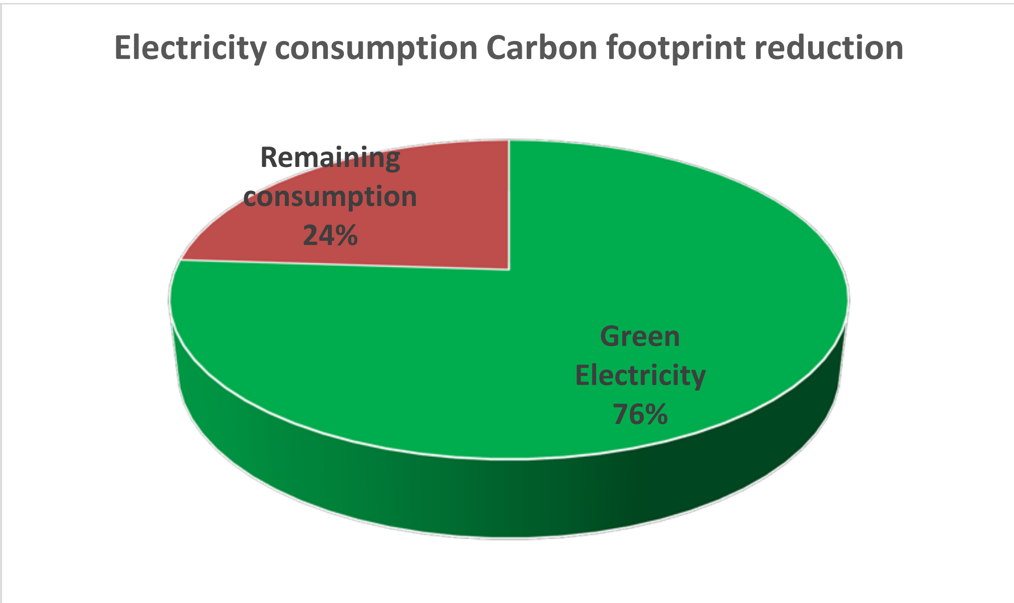
4.1. General terms

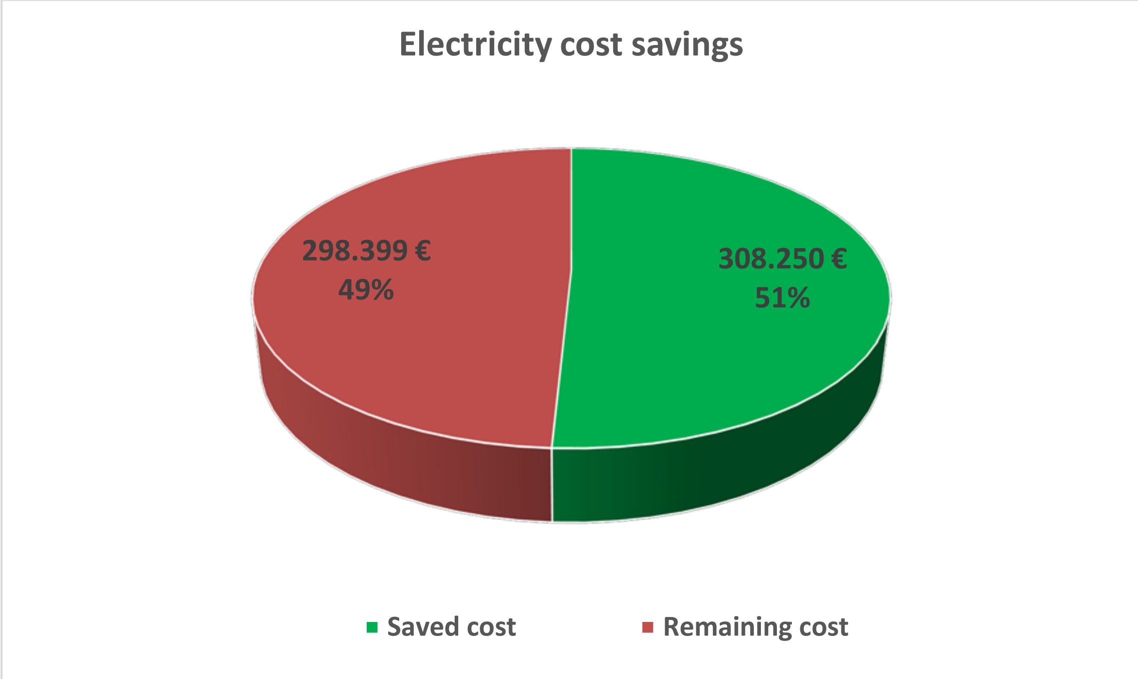
Energy communities can be founded and install photovoltaic systems in order to cover their energy consumption needs through virtual net metering of energy (N.4414/2016 και N.4513/2018).

In the case of Four Seasons Astir Palace Hotel Athens, virtual net-metering can be applied by compensating their energy consumption through the production of a PV park in the prefecture of Attica and/or neighboring prefectures. For that reason, an **energy community, non-profit organization of five members** (private VAT registries) must be established. Four Seasons Astir Palace Hotel Athens will participate among the member of the Energy Community. The PV park generation will offset the consumption of all community members.

In that way, Four Seasons Astir Palace Hotel Athens will reduce its carbo footprint by offsetting a large part of their electricity consumption by green electricity produced by the PV. At the same time, the company will achieve **significant cost savings** though the counterbalance of the electricity supplier's energy production tariff with the generation of the PV system while supporting the company's green profile. As a final step, the **certification of the carbon footprint reduction** can take place by a designated certification body so as the company can formally publish its green profile.

The contract for virtual net-metering has a duration of 25 years. The energy produced through the PV can be compensated by consumption within three (3) years.





4.2. Permitting procedure

Briefly, the steps for achieving project licensing are presented:

- Establishment of energy community as a non-profit organization of five members (five private VAT registries)
- Submission of grid connection application to HEDNO
- Environmental study preparation and permitting
- Issuance and acceptance of grid connection offer
- Drafting and signing of grid connection Agreement
- Drafting and signing of a Net-metering Agreement with the electricity supplier
- Installation of photovoltaic system
- Connection to the grid

5. ENERGY SAVINGS TARIFF

The tariff of generation compensation resulted from the supplier's electricity bills for the month of October 2020.

Further analysis needs to be performed in order to estimate the expected tariff as the generation cost variates according to billing from supplier and the wholesale market price of electricity.

Electricity bill analysis	**€/kWh**
Generation	0,0685
Distribution network	-
Transmission grid	-
Other charges	-
Other taxes	-
Total net-metering price	**0,0685**

6. Photovoltaic plant characteristics

The following table presents the basic features of the proposed photovoltaic plant:

PV plant features		
PV power	**3.0**	MWp
Annual PV generation	4,500,000	kWh
Annual savings (1st year)	308,250	€

The aforementioned estimations will be revised with the results of the detailed analysis.

The photovoltaic park, is assumed of installed capacity 3,0 MWp which is the maximum with the present institutional framework.

7. OUR PROPOSAL

Three options for the provision of services are provided as follows:

1. 100% self-financing
2. 70% loan and 30% participation
3. 70% loan and 8% initial participation and 8% buyout after eight years (8-year Energy Performance Contract)

All three options include the following services:

- Permitting procedure
- Establishment of energy community
- Engineering – Procurement – Construction and Commissioning of the park
- Operation and maintenance services
- Electrical grid connection fees

In addition, the 3rd option includes performance warranty and monitoring services.

Financial features	Self financing	Loan	EPC
CAPEX participation	2.135.000 €	640.000 €	170.000 €
Loan	0 €	1.495.000 €	1.494.500 €
Payback	8 years	5 years	10 years
Payout price (year 8)	0 €	0 €	170.000 €
NPV at year 8	-314.451 €	219.473 €	-74.073 €

The three options presented are formed on a **budgeting basis** and will be appropriately reformed with the actual figures after the detailed analysis.

The following are not included:

- Research for and purchase of field suitable for PV park installation, estimated at 45.000 sq.m. at a price of purchase of
- Insurance fees
- Loan issuance fees
- Due diligence – certification fees

8. ASSUMPTIONS

For our analysis the following assumptions were applied as presented in the table:

Installation Figures		
Installation Cost	2.135.000	€
Annual savings (1st year)	308.250	€
Annual maintenance cost	15.000	€
Nominal Power	3	MWp
Annual electricity production	4.500.000	kWh

Annual increase rate of electricity tariff	0,00%
Annual increase rate of maintenance cost	0,00%

Interest rate	**5,0%**
Debt % of total capex	**70%**
Duration (years)	**14**

9. FINANCIAL RESULTS

Concerning the financial results of the options the following indicators were used:

- Simple payback period
- IRR – Internal Rate of Return
- Net present value

The following table presents the results for the 25-year period:

Οικονομικά Αποτελέσματα πελάτη	Self financing	Loan	EPC
IRR (25-year)	12,2%	21,7%	21,9%
NPV (5%, 25-year)	4.548.925 €	1.712.977 €	1.284.094 €

Thanking you in advance for your interest we remain at your disposal for any further information

On behalf of ZEB S.A,

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ANNEX I: Cashflows of the three options

Απόδοση		1	0,97	0,965	0,96	0,955	0,95	0,945	0,94	0,935	0,93
Year	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Energy Savings		4.500.000 kWh	4.365.000 kWh	4.342.500 kWh	4.320.000 kWh	4.297.500 kWh	4.275.000 kWh	4.252.500 kWh	4.230.000 kWh	4.207.500 kWh	4.185.000 kWh
Cost savings		308.250 €	299.003 €	297.461 €	295.920 €	294.379 €	292.838 €	291.296 €	289.755 €	288.214 €	286.673 €
Loan		-148.699 €	-148.699 €	-148.699 €	-148.699 €	-148.699 €	-148.699 €	-148.699 €	-148.699 €	-148.699 €	-148.699 €
Maintenance		-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €
Self financing	-2.135.000 €	293.250 €	284.003 €	282.461 €	280.920 €	279.379 €	277.838 €	276.296 €	274.755 €	273.214 €	271.673 €
Debt 70%	-640.000 €	144.551 €	135.303 €	133.762 €	132.221 €	130.679 €	129.138 €	127.597 €	126.056 €	124.514 €	122.973 €
Energy Performance Contract	-170.000 €	15.413 €	14.950 €	14.873 €	14.796 €	14.719 €	14.642 €	14.565 €	-155.512 €	123.764 €	127.971 €

Απόδοση	0,925	0,92	0,915	0,91	0,905	0,9	0,895	0,89	0,885	0,88	0,875	0,87	0,865	0,86	0,855
Year	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**
Energy Savings	4.162.500 kWh	4.140.000 kWh	4.117.500 kWh	4.095.000 kWh	4.072.500 kWh	4.050.000 kWh	4.027.500 kWh	4.005.000 kWh	3.982.500 kWh	3.960.000 kWh	3.937.500 kWh	3.915.000 kWh	3.892.500 kWh	3.870.000 kWh	3.847.500 kWh
Cost savings	285.131 €	283.590 €	282.049 €	280.508 €	278.966 €	277.425 €	275.884 €	274.343 €	272.801 €	271.260 €	269.719 €	268.178 €	266.636 €	265.095 €	263.554 €

	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Loan	-148.699 €	-148.699 €	-148.699 €	-148.699 €	-148.699 €											
Maintenance	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €	-15.000 €
Self financing	271.673 €	270.131 €	268.590 €	267.049 €	265.508 €	263.966 €	262.425 €	260.884 €	259.343 €	257.801 €	256.260 €	254.719 €	253.178 €	251.636 €	250.095 €	248.554 €
Debt 70%	122.973 €	121.432 €	119.891 €	118.349 €	116.808 €	263.966 €	262.425 €	260.884 €	259.343 €	257.801 €	256.260 €	254.719 €	253.178 €	251.636 €	250.095 €	248.554 €
Energy Performance Co	127.971 €	132.177 €	136.384 €	140.591 €	144.798 €	263.966 €	262.425 €	260.884 €	259.343 €	257.801 €	256.260 €	254.719 €	253.178 €	251.636 €	250.095 €	248.554 €

ANNEX II: Energy Community Example

Παραδείγματα δυνητικής εφαρμογής στην ελληνική περίπτωση (3/13)

3.	ΕΚΟΙΝ-Επιχειρήσεις	ΦΩΤΟΒΟΛΤΑΙΚΟ – VIRTUAL NET METERING	Τουριστική Περιοχή - Νησί
Πέντε επιχειρήσεις (ξενοδοχεία, κλπ) δημιουργούν μία Ενεργειακή Κοινότητα προκειμένου να εγκαταστήσουν ένα φ/β σύστημα ή και μία α/γ σε ένα ακίνητο που μπορεί να βρίσκεται ακόμη και σε κάποια άλλη περιοχή και εφαρμόζουν **εικονικό ενεργειακό συμψηφισμό** της παραγόμενης ηλεκτρικής ενέργειας από το φ/β σύστημα ή την α/γ με τις καταναλώσεις των επιχειρήσεων, για να μειώσουν το ενεργειακό τους κόστους. Ενδεικτική Ισχύς: 500kW Ενδεικτικός Προϋπολογισμός: 500.000€ **Κίνητρα:** εικονικός ενεργειακός συμψηφισμός με καταναλώσεις μελών της ΕΚΟΙΝ.			

You may find the full legal provision of L4513/2018 concerning Energy Communities at the following link:

https://www.energia.gr/media/inlinepics/4513_2018EnergiakCommun.pdf